

5 Much success was made in reducing the number of cases of malaria between 2000 and 2015.

(a) Explain how malaria is transmitted.

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(b) Diagnostic test strips for malaria contain monoclonal antibodies. The test strips detect antigens produced by the pathogens that cause malaria.

Fig. 5.1 shows stages in the production of monoclonal antibodies. The information in three of these stages is incomplete.

Complete Fig. 5.1.

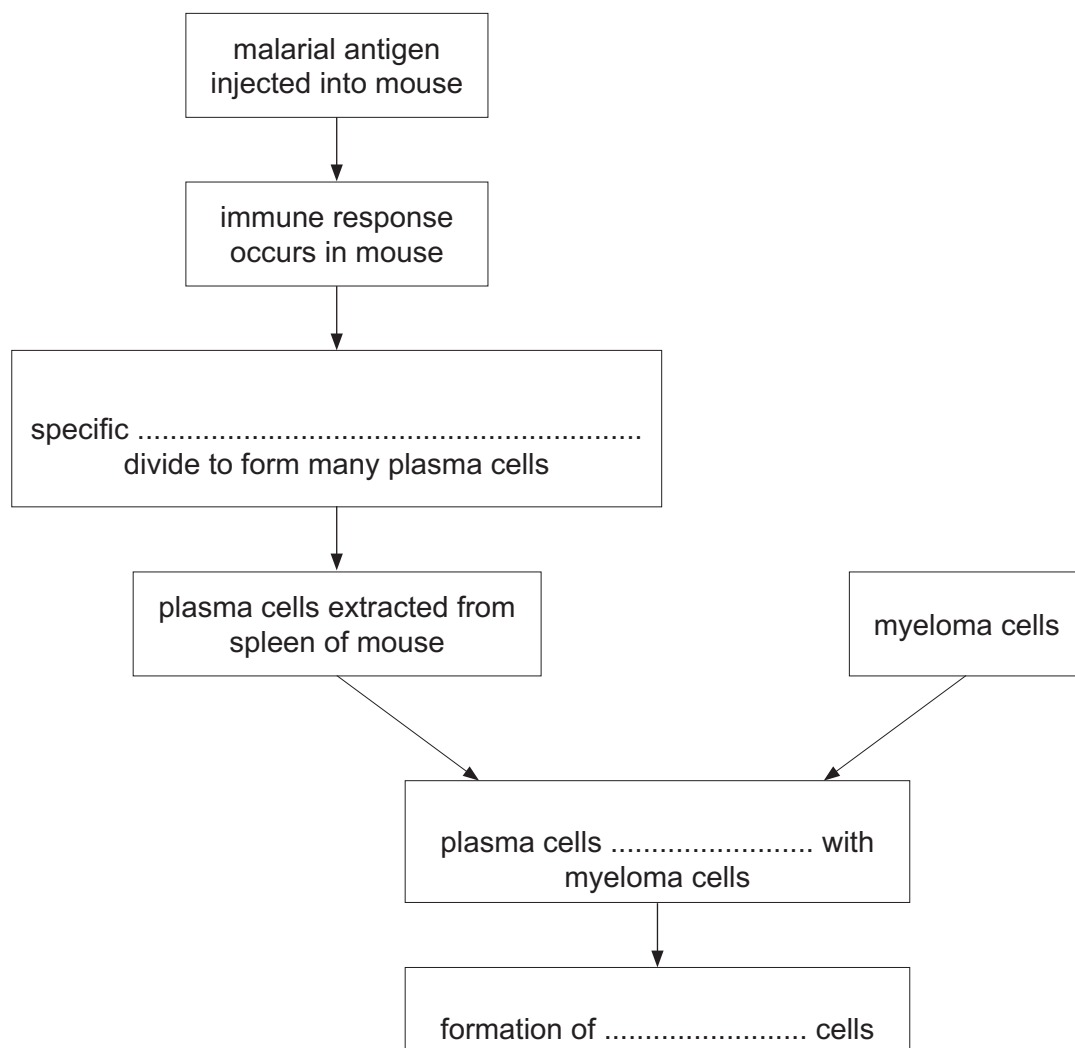


Fig. 5.1

(c) Fig. 5.2 shows two diagnostic test strips for malaria.

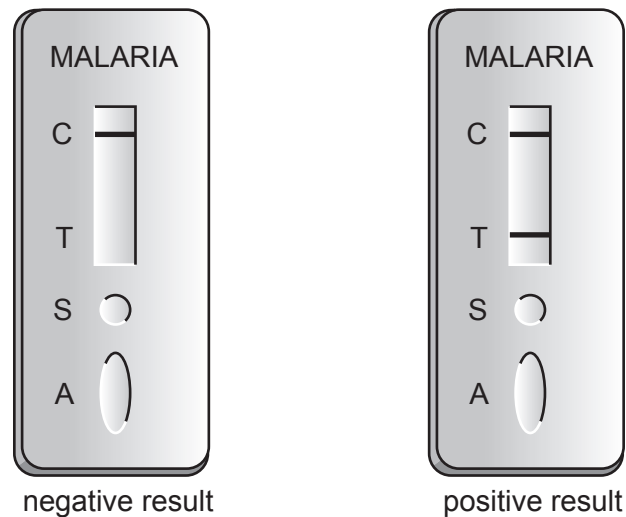


Fig. 5.2

- A sample of blood from a person suspected of having malaria is put into the well labelled **S**.
- A buffer solution is put into the well labelled **A**.
- The buffer solution moves the blood towards the results window.
- A line at position **C** indicates that the test is working correctly.
- A line at position **T** indicates a positive result for malaria.

State **three** advantages of using test strips for malaria, such as those shown in Fig. 5.2.

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(d) The highest number of cases of malaria occur in sub-Saharan Africa and South-East Asia.

Discuss the factors that determine the global distribution of malaria.

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